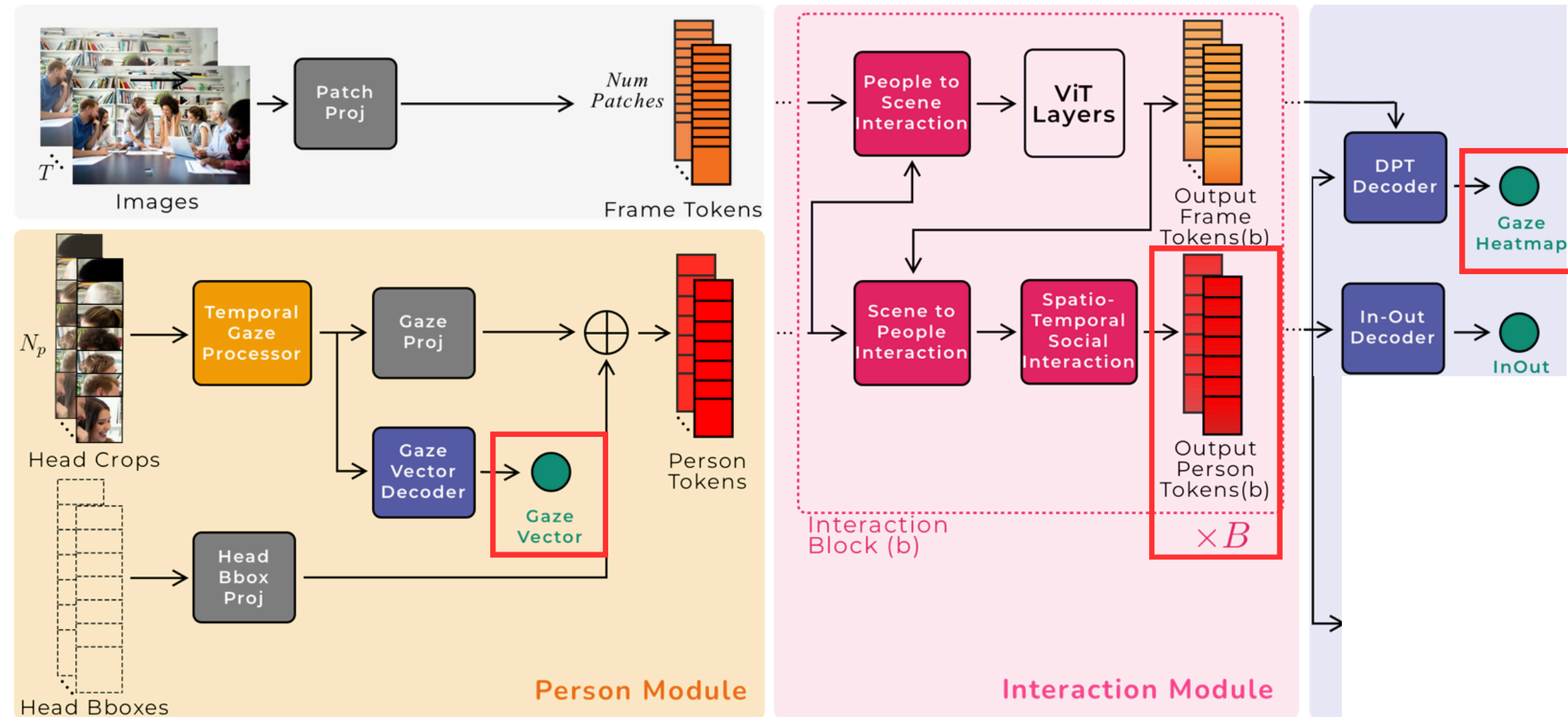


1. Gaze-Graph Mode

MTGS: Evidence Extractor ❄️



Extracted person gaze evidence :

1. person token h_i
2. gaze vector g_i
3. gaze heatmap H_i

1. Gaze-Graph Mode

Stage 1 : Node Initialization

1. Person Node $P_i = \text{LN}(h_i + \text{Cross-Attn}(h_i , h_{m_i}))$
 - gaze heatmap $\rightarrow$ 8x8 grid pooling $\rightarrow$ projection $\rightarrow h_{m_i}$
 - “나는 누구이고, 나는 어디를 바라보는가”
2. Region Node $R_k = \text{LN}(\text{PE}(\text{pred_anchor}_k) + \text{Cross-Attn}(\text{PE}(\text{pred_anchor}_k), h_k))$
 - “person k가 바라보는 위치이고, 그 주체가 person k이다”

Source / Target	P1	P2	P3	R1	R2	R3	Null
P1							
P2							
P3							

1. Gaze-Graph Mode

Stage 2 : Edge Initialization

Initial Edge $e^0_{i \rightarrow t} = \text{MLP}([\text{source} \parallel \text{target} \parallel \text{prior}_w[s_hm_{i \rightarrow t}, \text{align}_{i \rightarrow t}]])$

- $s_hm_{i \rightarrow t}$: heatmap-target overlap
- $\text{align}_{i \rightarrow t}$: gaze-target alignment (cosine)

→ $e_{i \rightarrow P_j}$: “i의 시선이 j의 머리를 향하는가?” – LAH/LAEO

$i \neq j$

→ $e_{i \rightarrow R_k}$: “i의 시선이 person k가 보는 곳을 향하는가?” – SA

$i \neq k$

Source / Target	P1	P2	P3	R1	R2	R3	Null
P1	–	e12	e13	–	e1R2	e1R3	e1N
P2	e21	–	e23	e2R1	–	e2R3	e2N
P3	e31	e32	–	e3R1	e3R2	–	e3N

1. Gaze-Graph Mode

Stage 3 : Dual-Role Edge Refinement

Step 1 – Row-wise outgoing attention

- source i 를 고정, i 에서 나가는 T 개 edge를 sequence로 두어 attention
- $[e_{i \rightarrow \text{target } 1} , \dots , e_{i \rightarrow \text{target } T}] \rightarrow \text{Transformer}_{\text{row}} \rightarrow [e'_{i \rightarrow \text{target } 1} , \dots , e'_{i \rightarrow \text{target } T}]$ (in-place)

Source / Target	P1	P2	P3	R1	R2	R3	Null
P1	–	e'_{12}	e'_{13}	–	e'_{1R2}	e'_{1R3}	e'_{1N}
P2	e'_{21}	–	e'_{23}	e'_{2R1}	–	e'_{2R3}	e'_{2N}
P3	e'_{31}	e'_{32}	–	e'_{3R1}	e'_{3R2}	–	e'_{3N}

1. Gaze-Graph Mode

Stage 3 : Dual-Role Edge Refinement

Step 2 – Column-wise incoming attention

- target t 를 고정, t 로 들어오는 N 개 edge를 sequence로 두어 attention
- $[e'_{\text{source } 1 \rightarrow t} , \dots , e'_{\text{source } N \rightarrow t}] \rightarrow \text{Transformer}_{\text{column}} \rightarrow [e''_{\text{source } 1 \rightarrow t} , \dots , e''_{\text{source } N \rightarrow t}]$ (in-place)

Source / Target	P1	P2	P3	R1	R2	R3	Null
P1	–	e''_{12}	e''_{13}	–	e''_{1R2}	e''_{1R3}	e''_{1N}
P2	e''_{21}	–	e''_{23}	e''_{2R1}	–	e''_{2R3}	e''_{2N}
P3	e''_{31}	e''_{32}	–	e''_{3R1}	e''_{3R2}	–	e''_{3N}

1. Gaze-Graph Mode

Stage 3 : Dual-Role Edge Refinement

Step 3 – Node Update

- $message_{src_i} = \text{mean over valid } t \text{ of } e_{i \rightarrow t}$ $\leftarrow$ i에서 나가는 모든 outgoing 엣지 평균
- $message_{tgt_t} = \text{mean over valid } i \text{ of } e_{i \rightarrow t}$ $\leftarrow$ t로 들어오는 모든 incoming 엣지 평균

- $node_{src_i} = \text{LN}(node_{src_i} + \text{MLP}([node_{src_i} \parallel message_{src_i}]))$ # 어디를 보는가
- $node_{tgt_t} = \text{LN}(node_{tgt_t} + \text{MLP}([node_{tgt_t} \parallel message_{tgt_t}]))$ # 누가 나를 보는가

Source / Target	P1	P2	P3	R1	R2	R3	Null
P1	–	e''_{12}	e''_{13}	–	e''_{1R2}	e''_{1R3}	e''_{1N}
P2	e''_{21}	–	e''_{23}	e''_{2R1}	–	e''_{2R3}	e''_{2N}
P3	e''_{31}	e''_{32}	–	e''_{3R1}	e''_{3R2}	–	e''_{3N}

1. Gaze-Graph Mode

Stage 3 : Dual-Role Edge Refinement

Step 4 – Edge Refresh

• $e^1_{i \rightarrow t} \leftarrow \text{LN}(e''_{i \rightarrow t} + \text{MLP}([e''_{i \rightarrow t} \parallel \text{node}_{\text{src}_i} \parallel \text{node}_{\text{tgt}_t}]))$

Step 1 ~ Step 4까지의 Dual-Role Edge Refinement를 L(=2)번 반복 진행

Source / Target	P1	P2	P3	R1	R2	R3	Null
P1	–	e12	e13	–	e1R2	e1R3	e1N
P2	e21	–	e23	e2R1	–	e2R3	e2N
P3	e31	e32	–	e3R1	e3R2	–	e3N

1. Gaze-Graph Mode

Stage 4 : Read-out

L(=2) iteration 0 이후 final edge $\hat{e}_{i \rightarrow t}$ 에 대하여, 공통 MLP 적용하여 logit 생성

- 1. $\text{mat}_{\text{LAH}}[i, j] = e_{i \rightarrow P_j} \text{ logit} \rightarrow \text{LAH}$ (i가 j의 머리를 본다)
- 2. $\text{mat}_{\text{SA}}[i, j] = e_{i \rightarrow R_j} \text{ logit} \rightarrow \text{SA}$ (i가 j가 보는 곳을 본다)
- 3. $\text{vec}_{\text{null}}[i] = e_{i \rightarrow \text{null}} \text{ logit} \rightarrow \text{out-of-frame}$

이후 각 scalar에 대해 1/0 BCE loss 걸침

Source / Target	P1	P2	P3	R1	R2	R3	Null
P1	-	e12	e13	-	e1R2	e1R3	e1N
P2	e21	-	e23	e2R1	-	e2R3	e2N
P3	e31	e32	-	e3R1	e3R2	-	e3N